Activities and experiences

Research

11 (Break)
105 hr/wk, 6 wk/yr

Researcher, Summer Science Program, Astrophysics, Preliminary Orbit Determination of NEA GF27 2015, Mentors: Dr Le, Dr. Mcneese, Dr. Rodruck
Learnt orbital mechanics & tensor/vector calculus. Applied Gauss' Method on NEA 2015 GF27. Achieved low error of 0.003%. Submitted results to IAU MPC.

Research

11, 12 (School)
17 hr/wk, 20 wk/yr
Will continue

Researcher, SiNRG UIUC, Optimizing learned H theta for low error in HRTF Spatial Audio Rendering, Mentor: Prof. Romit Roy Choudhury
Collected binaural data; estimated HRTFs via STFT/FFT + regularized least-squares; measured localization error; applied to biosignal spatial music.

Community service or volunteer work

9, 10, 11, 12 (Year)
38 hr/wk, 10 wk/yr
Will continue

Founder, The Lens Project, providing free & accessible eyecare to underserved communities, in rural West Bengal, India. Mentor: Dr. Seel
Screened 7k villagers; provided 1.7k lens surgeries, 3k lenses & 17 Hansen chairs. Served 1k leprosy patients; raised $14k. Built mobile cataract app.

Robotics

10, 11, 12 (Year)
20 hr/wk, 30 wk/yr

Developer, IISc, Sensorless Indoor Localization, Navigation & Control of Semi-Autonomous Wheelchair, Mentor: Prof. Pradipta Biswas, CPDM
Built voice-controlled wheelchair; used dead-reckoning localization, floorplan mapping, inverse kinematics & regression-based motor control in ROS

Research

PREVIEW

11, 12 (Year)
18 hr/wk, 15 wk/yr

Researcher, Medical AI, Diagnostic-Language Dependence and Robustness in Multimodal Medical LLMs, Mentor: Dr. Kiran Nijjer
Designed benchmark & perturbations; evaluated 5 LLMs on 657 cases on 29 conditions; ran controls/bootstraps; coauthored WMHS @ NeurIPS 2026 paper.

Music

9, 10, 11, 12 (Year)
12 hr/wk, 50 wk/yr
Will continue

Pianist, Natallia Piano Studio, Performed in 8 venues across Bangalore, India, Mentor: Professor Natallia Kapylova
Classically trained for 12 years. LCM Grade 8 High Achiever with distinction. Performed in Royal Albert Hall, London. Regular recital performer at BIC

Research

10, 11 (Year)
40 hr/wk, 20 wk/yr

Researcher, Jadavpur University, Wearable Athlete-Readiness Assessment Using Dermal Potential & ML, Mentor: Prof. Ratna Ghosh, Dept of Computer Eng
Collected athlete dermal-potential data; extracted biosignal features with ACF; trained EKF + Random Forest models; validated against real performance

Internship

10, 11 (Break)
14 hr/wk, 8 wk/yr

Intern, BrainSight.ai, ML Based Prognosis in Mild TBI, Diffusion MRI & Neuroimaging, Mentor: Dr. Rimjhim Aggarwal
Processed DTI data; generated whole-brain tractographs in MRtrix3/VoxelBox and extracted white-matter connectivity metrics toward an ML mTBI biomarker

Research

9, 10 (Year)
14 hr/wk, 10 wk/yr

Shadow Intern, Center for Brain Research, Indian Institute of Science (IISc), Mentor: Prof. Neelam Sinha
Youngest researcher at lab. Worked on MRTrix3 pipelines to generate adjacency matrices from schizophrenia dMRI/fMRI data for an ongoing project.

Student government or politics

9, 10, 11, 12 (Year)
8 hr/wk, 48 wk/yr

Secretary, Cyber Association, NPS HSR, Mentor: Ms. Sumati S and Ms. Nalina S.
Organized interschool Cyber Festival Cybernautica, 2023. Managed 300 participants and headed 2 events. Organized 25 algorithm workshops.

Responsibilities and circumstances

Household responsibilities

- Taking care of sick, disabled, and/or elderly members of my family or household

Circumstances

- Commuting 60 minutes or more to and from school each day